

*index, i, j, n, m*    *indices*  
*label, l, k*        *field label*  
termvar, x  
basetype, A

|           |                                                                                                                                                                             |                                                                                                                                                                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| terminals | $::=$<br>$\rightarrow$<br>$\rightarrow$<br>$\Rightarrow$<br>$\lambda$<br>$\mapsto$<br>$\vdash$<br>$\emptyset$<br>$\times$<br>$<:$<br>$\langle$<br>$\rangle$                 |                                                                                                                                                                                                                                               |
| labels    | $::=$<br>$l_1 .. l_n$                                                                                                                                                       |                                                                                                                                                                                                                                               |
| $\Gamma$  | $::=$<br>$\emptyset$<br>$\Gamma, x : T$                                                                                                                                     | <i>contexts:</i><br><i>empty context</i><br><i>term variable binding</i>                                                                                                                                                                      |
| formula   | $::=$<br>judgement<br>$formula_1 .. formula_n$<br>$j \text{ INDEXES } t_1 .. t_n$<br>$j \text{ INDEXES } T_1 .. T_n$<br>DISTINCT $l_1 .. l_n$<br>$x : T \in \Gamma$         |                                                                                                                                                                                                                                               |
| $T, S, U$ | $::=$<br>Bool<br>Nat<br>$T \rightarrow T'$<br>$(T)$<br>A<br>Unit<br>$T_1 \times T_2$<br>$T_1 + T_2$<br>$\{ T_i^{i \in 1..n} \}$<br>$\langle l_i : T_i^{i \in 1..n} \rangle$ | <i>types:</i><br><i>type of booleans</i><br><i>type of natural numbers</i><br><i>type of functions</i><br>M<br><i>base type</i><br><i>unit type</i><br><i>product type</i><br><i>sum type</i><br><i>tuple type</i><br><i>type of variants</i> |
| t         | $::=$<br>$(t)$<br>true<br>false<br>if $t_1$ then $t_2$ else $t_3$<br>0<br>succ t<br>pred t<br>iszero t<br>x                                                                 | <i>terms:</i><br>M<br><i>constant true</i><br><i>constant false</i><br><i>conditional</i><br><i>constant zero</i><br><i>successor</i><br><i>predecessor</i><br><i>zero test</i><br><i>variable</i>                                            |

|             |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                              |                                                                                                                                                                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | $\lambda x : T.t$<br>$t_1 t_2$<br>$[x \mapsto t']t$<br>$\text{unit}$<br>$t_1; t_2$<br>$t \text{ as } T$<br>$\text{let } x = t \text{ in } t'$<br>$\{t_1, t_2\}$<br>$t.1$<br>$t.2$<br>$\text{inl } t$<br>$\text{inr } t$<br>$\text{case } t \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2$ | $\text{bind } x \text{ in } t$<br><br>$M$<br><br><br>$\text{bind } x \text{ in } t'$<br><br><br><br>$\text{bind } x_1 \text{ in } t_1$<br>$\text{bind } x_2 \text{ in } t_2$ | <i>abstraction</i><br><i>application</i><br><br><br><i>constant unit</i><br><i>sequencing</i><br><i>ascription</i><br><i>let binding</i><br><i>pair</i><br><i>first projection</i><br><i>second projection</i><br><i>tagging (left)</i><br><i>tagging (right)</i><br><i>case</i>         |
|             | $\text{fix } t$<br>$\{t_i\}_{i \in 1..n}$<br>$t.i$<br>$\langle l = t \rangle \text{ as } T$<br>$\text{case } t \text{ of } C_i\_{i \in 1..n}$                                                                                                                                                                                |                                                                                                                                                                              | <i>fixed point of t</i><br><i>tuple</i><br><i>projection</i><br><i>tagging</i><br><i>case</i>                                                                                                                                                                                            |
| C           | $::=$<br>$\langle l = x \rangle \Rightarrow t$                                                                                                                                                                                                                                                                               | $\text{bind } x \text{ in } t$                                                                                                                                               |                                                                                                                                                                                                                                                                                          |
| v           | $::=$<br>$\text{true}$<br>$\text{false}$<br>$0$<br>$\text{succ } v$<br>$\lambda x : T.t$<br>$\text{unit}$<br>$\{v_1, v_2\}$<br>$\text{inl } v$<br>$\text{inr } v$<br>$\{v_i\}_{i \in 1..n}$<br>$\langle l = v \rangle \text{ as } T$                                                                                         | <br><br><br><br>$\text{bind } x \text{ in } t$<br><br><br><br><br><br>                                                                                                       | <i>values:</i><br><i>true value</i><br><i>false value</i><br><i>zero value</i><br><i>successor value</i><br><i>abstraction value</i><br><i>unit value</i><br><i>pair value</i><br><i>tagged value (left)</i><br><i>tagged value (right)</i><br><i>tuple value</i><br><i>tagged value</i> |
| Jop         | $::=$<br>$t \longrightarrow t'$                                                                                                                                                                                                                                                                                              |                                                                                                                                                                              | <i>Evaluation</i>                                                                                                                                                                                                                                                                        |
| Jtype       | $::=$<br>$\Gamma \vdash t : T$                                                                                                                                                                                                                                                                                               |                                                                                                                                                                              | <i>Typing</i>                                                                                                                                                                                                                                                                            |
| judgement   | $::=$<br>$\text{Jop}$<br>$\text{Jtype}$                                                                                                                                                                                                                                                                                      |                                                                                                                                                                              |                                                                                                                                                                                                                                                                                          |
| user_syntax | $::=$<br>$\text{index}$<br>$\text{label}$                                                                                                                                                                                                                                                                                    |                                                                                                                                                                              |                                                                                                                                                                                                                                                                                          |

termvar  
 basetype  
 terminals  
 labels  
 $\Gamma$   
 formula  
 $T$   
 $t$   
 $C$   
 $v$

$t \longrightarrow t'$     *Evaluation*

$$\begin{array}{c}
 \frac{}{\text{if true then } t_2 \text{ else } t_3 \longrightarrow t_2} \quad \text{E\_IFTRUE} \\
 \frac{}{\text{if false then } t_2 \text{ else } t_3 \longrightarrow t_3} \quad \text{E\_IFFALSE} \\
 \frac{t_1 \longrightarrow t'_1}{\text{if } t_1 \text{ then } t_2 \text{ else } t_3 \longrightarrow \text{if } t'_1 \text{ then } t_2 \text{ else } t_3} \quad \text{E\_IF} \\
 \frac{t_1 \longrightarrow t'_1}{\text{succ } t_1 \longrightarrow \text{succ } t'_1} \quad \text{E\_SUCC} \\
 \frac{}{\text{pred } 0 \longrightarrow 0} \quad \text{E\_PREDZERO} \\
 \frac{}{\text{pred } (\text{succ } v_1) \longrightarrow v_1} \quad \text{E\_PREDSUCC} \\
 \frac{t_1 \longrightarrow t'_1}{\text{pred } t_1 \longrightarrow \text{pred } t'_1} \quad \text{E\_PRED} \\
 \frac{}{\text{iszero } 0 \longrightarrow \text{true}} \quad \text{E\_ISZEROZERO} \\
 \frac{}{\text{iszero } (\text{succ } v_1) \longrightarrow \text{false}} \quad \text{E\_ISZEROSUCC} \\
 \frac{t_1 \longrightarrow t'_1}{\text{iszero } t_1 \longrightarrow \text{iszero } t'_1} \quad \text{E\_ISZERO} \\
 \frac{t_1 \longrightarrow t'_1}{t_1 t_2 \longrightarrow t'_1 t_2} \quad \text{E\_APP1} \\
 \frac{t_2 \longrightarrow t'_2}{v_1 t_2 \longrightarrow v_1 t'_2} \quad \text{E\_APP2} \\
 \frac{}{(\lambda x : T. t_{12}) v_2 \longrightarrow [x \mapsto v_2] t_{12}} \quad \text{E\_APPABS} \\
 \frac{t_1 \longrightarrow t'_1}{t_1; t_2 \longrightarrow t'_1; t_2} \quad \text{E\_SEQ} \\
 \frac{}{\text{unit}; t_2 \longrightarrow t_2} \quad \text{E\_SEQNEXT} \\
 \frac{}{v_1 \text{ as } T \longrightarrow v_1} \quad \text{E\_ASCRIBE} \\
 \frac{t_1 \longrightarrow t'_1}{t_1 \text{ as } T \longrightarrow t'_1 \text{ as } T} \quad \text{E\_ASCRIBE1}
 \end{array}$$

$$\begin{array}{c}
\frac{}{\text{let } x = v_1 \text{ in } t_2 \longrightarrow [x \mapsto v_1]t_2} \text{E\_LETV} \\
\frac{t_1 \longrightarrow t'_1}{\text{let } x = t_1 \text{ in } t_2 \longrightarrow \text{let } x = t'_1 \text{ in } t_2} \text{E\_LET} \\
\frac{}{\{v_1, v_2\}.1 \longrightarrow v_1} \text{E\_PAIRBETA1} \\
\frac{}{\{v_1, v_2\}.2 \longrightarrow v_2} \text{E\_PAIRBETA2} \\
\frac{t_1 \longrightarrow t'_1}{t_1.1 \longrightarrow t'_1.1} \text{E\_PROJ1} \\
\frac{t_1 \longrightarrow t'_1}{t_1.2 \longrightarrow t'_1.2} \text{E\_PROJ2} \\
\frac{t_1 \longrightarrow t'_1}{\{t_1, t_2\} \longrightarrow \{t'_1, t_2\}} \text{E\_PAIR1} \\
\frac{t_2 \longrightarrow t'_2}{\{v_1, t_2\} \longrightarrow \{v_1, t'_2\}} \text{E\_PAIR2} \\
\frac{}{\text{case } (\text{inl } v_0) \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 \longrightarrow [x_1 \mapsto v_0]t_1} \text{E\_CASEINL} \\
\frac{}{\text{case } (\text{inr } v_0) \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 \longrightarrow [x_2 \mapsto v_0]t_2} \text{E\_CASEINR} \\
\frac{t_0 \longrightarrow t'_0}{\text{case } t_0 \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 \longrightarrow \text{case } t'_0 \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2} \text{E\_CASESM} \\
\frac{t_1 \longrightarrow t'_1}{\text{inl } t_1 \longrightarrow \text{inl } t'_1} \text{E\_INL} \\
\frac{t_1 \longrightarrow t'_1}{\text{inr } t_1 \longrightarrow \text{inr } t'_1} \text{E\_INR} \\
\frac{}{\text{fix } (\lambda x : T_1. t_2) \longrightarrow [x \mapsto (\text{fix } (\lambda x : T_1. t_2))]]t_2} \text{E\_FIXBETA} \\
\frac{t_1 \longrightarrow t'_1}{\text{fix } t_1 \longrightarrow \text{fix } t'_1} \text{E\_FIX} \\
\frac{j \text{ INDEXES } v_i^{i \in 1..n}}{\{v_i^{i \in 1..n}\}.j \longrightarrow v_j} \text{E\_PROJTUPLE} \\
\frac{t_1 \longrightarrow t'_1}{t_1.i \longrightarrow t'_1.i} \text{E\_PROJTP} \\
\frac{t \longrightarrow t'}{\{v_i^{i \in 1..m}, t, t_j^{j \in 1..n}\} \longrightarrow \{v_i^{i \in 1..m}, t', t_j^{j \in 1..n}\}} \text{E\_TUPLE} \\
\frac{j \text{ INDEXES } t_i^{i \in 1..n}}{\text{case } (\langle l_j = v \rangle \text{ as } T) \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n} \longrightarrow [x_j \mapsto v]t_j} \text{E\_CASEVARIANT} \\
\frac{t \longrightarrow t'}{\text{case } t \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n} \longrightarrow \text{case } t' \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n}} \text{E\_CASE} \\
\frac{t_i \longrightarrow t'_i}{\langle l_i = t_i \rangle \text{ as } T \longrightarrow \langle l_i = t'_i \rangle \text{ as } T} \text{E\_VARIANT}
\end{array}$$

|                                                                                                                              |              |
|------------------------------------------------------------------------------------------------------------------------------|--------------|
| $\overline{\Gamma \vdash \text{true} : \text{Bool}}$                                                                         | $T\_TRUE$    |
| $\overline{\Gamma \vdash \text{false} : \text{Bool}}$                                                                        | $T\_FALSE$   |
| $\Gamma \vdash t_1 : \text{Bool}$<br>$\Gamma \vdash t_2 : T$<br>$\Gamma \vdash t_3 : T$                                      | $T\_IF$      |
| $\overline{\Gamma \vdash 0 : \text{Nat}}$                                                                                    | $T\_ZERO$    |
| $\Gamma \vdash t_1 : \text{Nat}$<br>$\Gamma \vdash \text{succ } t_1 : \text{Nat}$                                            | $T\_SUCC$    |
| $\Gamma \vdash t_1 : \text{Nat}$<br>$\Gamma \vdash \text{pred } t_1 : \text{Nat}$                                            | $T\_PRED$    |
| $\Gamma \vdash t_1 : \text{Nat}$<br>$\Gamma \vdash \text{iszero } t_1 : \text{Bool}$                                         | $T\_ISZERO$  |
| $\frac{x : T \in \Gamma}{\Gamma \vdash x : T}$                                                                               | $T\_VAR$     |
| $\Gamma, x : T_1 \vdash t_2 : T_2$<br>$\Gamma \vdash \lambda x : T_1. t_2 : T_1 \rightarrow T_2$                             | $T\_ABS$     |
| $\Gamma \vdash t_1 : T_{11} \rightarrow T_{12}$<br>$\Gamma \vdash t_2 : T_{11}$<br>$\Gamma \vdash t_1 t_2 : T_{12}$          | $T\_APP$     |
| $\overline{\Gamma \vdash \text{unit} : \text{Unit}}$                                                                         | $T\_UNIT$    |
| $\Gamma \vdash t_1 : \text{Unit}$<br>$\Gamma \vdash t_2 : T_2$<br>$\Gamma \vdash t_1; t_2 : T_2$                             | $T\_SEQ$     |
| $\Gamma \vdash t_1 : T$<br>$\Gamma \vdash t_1 \text{ as } T : T$                                                             | $T\_ASCRIBE$ |
| $\Gamma \vdash t_1 : T_1$<br>$\Gamma, x : T_1 \vdash t_2 : T_2$<br>$\Gamma \vdash \text{let } x = t_1 \text{ in } t_2 : T_2$ | $T\_LET$     |
| $\Gamma \vdash t_1 : T_1$<br>$\Gamma \vdash t_2 : T_2$<br>$\Gamma \vdash \{t_1, t_2\} : T_1 \times T_2$                      | $T\_PAIR$    |
| $\Gamma \vdash t_1 : T_1 \times T_2$<br>$\Gamma \vdash t_1.1 : T_1$                                                          | $T\_PROJ1$   |
| $\Gamma \vdash t_1 : T_1 \times T_2$<br>$\Gamma \vdash t_1.2 : T_2$                                                          | $T\_PROJ2$   |
| $\Gamma \vdash t_1 : T_1$<br>$\Gamma \vdash \text{inl } t_1 : T_1 + T_2$                                                     | $T\_INL$     |
| $\Gamma \vdash t_1 : T_2$<br>$\Gamma \vdash \text{inr } t_1 : T_1 + T_2$                                                     | $T\_INR$     |

$$\begin{array}{c}
\frac{\Gamma \vdash t_0 : T_1 + T_2 \quad \Gamma, x_1 : T_1 \vdash t_1 : T \quad \Gamma, x_2 : T_2 \vdash t_2 : T}{\Gamma \vdash \text{case } t_0 \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 : T} \quad \text{T\_CASESM} \\
\\
\frac{\Gamma \vdash t_1 : T_1 \rightarrow T_1}{\Gamma \vdash \text{fix } t_1 : T_1} \quad \text{T\_FIX} \\
\\
\frac{\Gamma \vdash t_i : T_i^{i \in 1..n}}{\Gamma \vdash \{t_i^{i \in 1..n}\} : \{T_i^{i \in 1..n}\}} \quad \text{T\_TUPLE} \\
\\
\frac{\Gamma \vdash t_1 : \{T_i^{i \in 1..n}\} \quad j \text{ INDEXES } T_i^{i \in 1..n}}{\Gamma \vdash t_1.j : T_j} \quad \text{T\_PROJTP} \\
\\
\frac{\Gamma \vdash t : T_j \quad j \text{ INDEXES } T_i^{i \in 1..n}}{\Gamma \vdash \langle l_j = t \rangle \text{ as } \langle l_i : T_i^{i \in 1..n} \rangle : \langle l_i : T_i^{i \in 1..n} \rangle} \quad \text{T\_VARIANT} \\
\\
\frac{\Gamma \vdash t : \langle l_i : T_i^{i \in 1..n} \rangle \quad \Gamma, x_i : T_i \vdash t_i : T^{i \in 1..n} \quad \text{DISTINCT } l_i^{i \in 1..n}}{\Gamma \vdash \text{case } t \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n} : T} \quad \text{T\_CASE}
\end{array}$$

Definition rules: 63 good 0 bad  
 Definition rule clauses: 119 good 0 bad